Dictionary

Facility: Represents a railway facility such as a station, depot, or yard.

Attribute	Type	Description
ID (PK)	number(38)	Unique identifier of the facility.
Name	varchar2(100)	Name of the facility (e.g., “São Romão”).
TypeID (FK)	number(38)	References **FacilityType**. Indicates the facility’s type (station, depot, etc.).
Address	varchar2(100)	Physical address of the facility.
City	varchar2(100)	City where the facility is located.
Capacity	number(10)	Maximum capacity (e.g., trains, wagons, or passengers).
Electrified	char(3)	Indicates if the facility is electrified (‘YES’/‘NO’).
OwnerShortName (FK)	varchar2(50)	References short name of the owner who is responsible for managing or maintaining the facility.

FacilityType: Defines the category of a facility (e.g., station, depot, terminal).

Attribute	Data Type	Description
FacilityTypeID (PK)	number(38)	Unique identifier for each facility type.
Name	varchar2(100)	Name of the type (e.g., “Station”, “Depot”).

BuildingType: Defines the category of a building (e.g., warehouse, silo, office) used to classify buildings within facilities.

Attribute	Data Type	Description
ID (PK)	number(10)	Unique building type identifier.
Type	varchar2(100)	Type of building.

Facility_Building: Associative entity that records how many buildings of a specific building type exist within a given facility.

Attribute	Data Type	Description
BuildingTypeID (FK)	number(10)	References building type.
FacilityID (FK)	number(38)	References facility.
Number	number(10)	Number of such buildings in the facility (count).

Line: Represents a railway line connecting two facilities.

Attribute	Type	Description
ID (PK)	number(38)	Unique identifier for the line.
Name	varchar2(100)	Line name or designation.
OwnerShortName (FK)	varchar2(50)	References short name of the owner who is responsible for the line’s maintenance and operation.
StartID (FK)	number(38)	References **Facility** where the line begins.
EndID (FK)	number(38)	References **Facility** where the line ends.
GaugeName (FK)	varchar2(100)	References Gauge entity.

Segment: Defines a segment or section of a railway line.

Attribute	Type	Description
ID (PK)	number(38)	Unique identifier for the segment.
Order	number(10)	The segment's order in the line (1st, 2nd, etc.).
Electrified	char(3)	Indicates if the segment is electrified ('YES'/'NO').
Max_weight	float(38)	Maximum allowed train weight on this segment (kg/m).
Length	number(38)	Segment length (m).
NumberOfTracks	number(10)	Number of parallel tracks in the segment.
SpeedLimit	number(10)	Maximum allowed speed on the segment.

Line_Segment: Associative entity that links lines and segments, showing which segments belong to which line and in what structure.

Attribute	Type	Description
LineID (PK/FK)	number(38)	Specific railway line to which the segment belongs.
SegmentID (PK/FK)	number(38)	Uniquely identifies each segment.

Owner: Represents an organization that owns facilities or lines.

Attribute	Type	Description
ShortName (PK)	varchar2(50)	Abbreviated company name.
Name	varchar2(100)	Full name of the owning company.
VATnumber	varchar2(15)	VAT number of the company.

Train: Represents an individual train service operating on a specific schedule.

Attribute	Type	Description
ID (PK)	number(38)	Unique identifier for each train instance.
MaxLength	number(10)	Maximum allowed length of the train.

Planned_Route: Represents the planned sequence of lines and timings that a train is scheduled to follow before execution.

Attribute	Type	Description
TrainID (FK)	number(38)	Train assigned to the route.
LineID (FK)	number(38)	Line used in the route.
Direction	varchar2(100)	Direction of travel.
ArrivalTime	date	Planned arrival time.
Order	number(10)	Order of the line in the route.
DepartureTime	date	Planned departure time.

ActualRoute: Represents the actual executed route of a train, as opposed to the planned route.

Attribute	Type	Description
TrainID (PK, FK)	number(38)	Identifies the train whose actual route is recorded.
LineID (PK, FK)	number(38)	Identifies the line used in the actual route.

Direction	varchar2(100)	Direction of travel on the line (e.g., forward / reverse).
DepartureTime	date	Actual departure time of the train from this line.

ActualRouteArrival: Stores arrival details of a train at a specific point during its actual route.

Attribute	Type	Description
TrainID (PK, FK)	number(38)	Identifies the train arriving.
LineID (PK, FK)	number(38)	Identifies the line where the arrival occurs.
ArrivalTime	date	Actual arrival time of the train on the line.

Siding: Represents a secondary track connected to a segment, used for parking, waiting or overtaking trains.

Attribute	Type	Description
ID (PK)	number(38)	Unique identifier of the siding.
Position	number(10)	Position of the siding along the segment.
Length	number(10)	Length of the siding in meters.
SegmentID (FK)	number(38)	Identifies the segment to which the siding belongs.

Locomotive_Train: Links locomotives to trains, showing which locomotives are assigned to which train.

Attribute	Type	Description
LocomotiveLocomotiveNumber (PK/FK)	number(38)	Identifies the locomotive assigned to a train.
TrainID (PK/FK)	number(38)	Indicates which train a locomotive is linked to.

Freight: Represents a shipment transported by a train between two locations.

Attribute	Type	Description
ID (PK)	number(38)	Unique identifier of a freight shipment.
Freight	number(10)	A numeric value representing the unique identifier assigned to each freight record.
Weight	number(10)	Specifies the weight of the freight shipment in suitable units.
StartID (FK)	number(38)	Identifier of the departure facility or location for the freight.
EndID (FK)	number(38)	Identifier of the destination facility or location for the freight.
LoadTime	date	Loading time.

Train_Freight: An associative entity that links each freight record to the specific train responsible for transporting it.

Attribute	Type	Description
TrainID (PK/FK)	number(38)	Refers to the unique identifier of a train.
FreightID (PK/FK)	number(38)	Refers to the unique identifier of a freight item.

Train_Line: Links trains to the lines they operate on, along with arrival and departure times.

Attribute	Type	Description
TrainID (PK/FK)	number(38)	References the train that operates on the given line segment.
LineID (PK/FK)	number(38)	References the railway line involved in the train's operation.
ArrivalTime	date	The scheduled time when the train arrives at the line's endpoint or station.
DepartureTime	date	The scheduled time when the train leaves the start point of the line.

Wagon_Freight: Links wagons to freight units, indicating which wagons carry which specific freight.

Attribute	Type	Description
WagonWagonNumber (PK/FK)	varchar2(50)	Identifies the wagon that is assigned to carry a specific freight.
FreightID (PK/FK)	number(38)	Identifies the freight unit transported in the wagon.

Operator: Represents a person or entity responsible for operating locomotives or wagons.

Attribute	Type	Description
ShortName (PK)	varchar2(50)	Abbreviated operator name.
Name	varchar2(100)	Full name of the operator company.
VATnumber	varchar2(15)	VAT number of the company.

Gauge: Represents the track width standard used by locomotives, wagons, lines or facilities.

Attribute	Type	Description
Name (PK)	varchar2(100)	Unique name of the gauge.
Gauge	number(10)	Width of the track in millimeters (e.g., 1435 for standard).

Wagon: Represents a railway wagon or freight car.

Attribute	Type	Description
WagonNumber (PK)	number(38)	Unique identifier for each wagon.
ModelID (FK)	number(100)	References **WagonModel** that defines its characteristics.
ServiceYear	number(4)	Year the wagon entered service.
OperatorShortName (FK)	varchar2(50)	References short name of the operator managing the wagon.

WagonType: Defines categories of wagons.

Attribute	Type	Description
ID (PK)	number(38)	Unique identifier for wagon type.
Name	varchar2(100)	Name of the type (e.g., Cereal wagon, Container wagon).

WagonModel: Defines the technical specifications of wagon models.

Attribute	Type	Description
ModelNumber (PK)	number(38)	Unique identifier for the wagon model.
Model	varchar2(100)	Model name or code.
Maker	varchar2(100)	Manufacturer of the wagon.
NumberBogies	number(1)	Number of bogies (wheel assemblies).
Bogies	varchar2(100)	Description or type of bogie configuration.
Length	number(10)	Wagon length (m).
Width	float(10)	Wagon width (m).
Height	float(10)	Wagon height (m).
Weight	float(10)	Total empty wagon weight (N).
MaxSpeed	number(10)	Maximum allowed speed.
Payload	float(10)	Maximum load capacity.
Volume	float(10)	Internal volume (m^3).
TypeID (FK)	number(38)	References **WagonType** to define wagon category.

Gauge_WagonModel: Connects wagon models to the gauges they are compatible with.

Attribute	Type	Description
GaugeName (PK/FK)	varchar2(100)	Indicates the rail gauge for which the wagon model is compatible.
WagonModelModelNumber (PK/FK)	number(38)	Identifies the specific wagon model associated with a gauge.

FreightWagonUnload: Represents the unloading event of a wagon carrying freight, recording when a wagon unloads its assigned freight.

Attribute	Type	Description
WagonFreightWagonNumber (PK, FK)	varchar2(50)	Identifies the wagon involved in the freight unloading.
WagonFreightFreightID (PK, FK)	number(38)	Identifies the freight being unloaded.
UnloadTime	date	Date and time when the freight was unloaded from the wagon.

Locomotive: Represents an individual locomotive in service.

Attribute	Type	Description
LocomotiveNumber (PK)	number(38)	Unique identifier (fleet number) for each locomotive.
Name	varchar2(100)	Locomotive name or designation.
ModelID (FK)	number(38)	References **LocomotiveModel** that defines its specifications.
Service	number(4)	Service year or operational year.
OperatorShortName (FK)	varchar2(50)	References short name of the operator managing the locomotive.

LocomotiveModel: Defines a locomotive model's technical characteristics.

Attribute	Type	Description
ID (PK)	number(38)	Unique identifier for the locomotive model.
Make	varchar2(100)	Manufacturer of the locomotive
Model	varchar2(100)	Model name or number.
NumberBogies	number(10)	Number of bogies (wheel assemblies).
Bogies	varchar2(100)	Description of bogie configuration.
Power	number(10)	Engine power rating (kW).
Length	float(10)	Locomotive length (m).
Width	float(10)	Locomotive width (m).
Height	float(10)	Locomotive height (m).
Weight	number(10)	Total locomotive weight (N).
MaxSpeed	number(10)	Maximum operational speed.
OperationalSpeed	float(10)	Regular service speed.
Traction	number(10)	Force of traction (N).
DieselTypeID (FK)	number(38)	References **Diesel_locomotive** if diesel-powered.
ElectricTypeID (FK)	number(38)	References **Electric_locomotive** if electric-powered.

LocomotiveModel_Gauge: Connects locomotive models to the gauges they can operate on.

Attribute	Type	Description
LocomotiveModelID (PK/FK)	number(38)	Identifies the locomotive model connected to a specific gauge.
GaugeName (PK/FK)	varchar2(100)	Indicates the gauge on which the locomotive model is capable of operating.

Diesel_locomotive: Represents specific data for diesel locomotive models.

Attribute	Type	Description
DieselTypeID (PK)	number(38)	Unique identifier for the diesel locomotive type.
Fuel_capacity	number(38)	Fuel tank capacity, combustivel (l).

Electric_locomotive: Represents specific data for electric locomotive models.

Attribute	Type	Description
ElectricTypeID (PK)	number(38)	Unique identifier for the electric locomotive type.
Voltage	number(10)	Operating voltage (V).
Frequency	number(10)	Frequency (Hz).

Glossary

General Concepts:

Term	Definition / Explanation
DBMS (Database Management System)	A software system that manages the creation, organization, and manipulation of a database. Examples: MySQL, PostgreSQL, Oracle.
Entity	A real-world object or concept represented as a table in the database (e.g., Facility, Line, Wagon).
Attribute	A property or characteristic of an entity (e.g., Name, ID, Capacity). Each attribute corresponds to a column in a table.
Primary Key (PK)	A unique identifier for each record in a table (e.g., ID, LocomotiveNumber).
Foreign Key (FK)	An attribute that creates a relationship between two entities by referencing another entity's primary key.
Relationship	A connection between entities (e.g., each Line connects two Facilities).
Cardinality	Defines how many instances of one entity relate to another (e.g., one Line can have many Segments).
Normalization	The process of structuring a database to reduce redundancy and improve data integrity.
VARCHAR(n)	Variable-length character data type; stores text up to *n* characters.
NUMBER(n)	Numeric data type; stores numeric values up to *n* digits.
FLOAT(n)	Floating-point data type; stores decimal numbers with precision.
CHAR(3)	Fixed-length character field (used for short codes like "Yes/No" or "Y/N").
U (Unique)	Constraint ensuring that all values in a column are distinct.
N (Nullable)	Indicates that the field can be left empty (NULL).

Relationships:

Relationship	Description
Facility – FacilityType	Each facility has one type. A type can be shared by multiple facilities.
Facility – Owner	Each facility is owned by one Owner.
Line – Facility	Each line connects two facilities (start and end).
Line – Segment	A line consists of one or many segments, each segment can be part of multiple lines.
Line – Train	A train can operate on multiple lines and each line can host multiple trains.
Line – Owner	A line has one owner responsible for maintenance.
Line – Gauge	Each line has one track gauge.
Segment – Siding	A segment can have zero or more sidings, each siding belongs to one segment.
Wagon – WagonModel	Each wagon has one model type.
WagonModel – WagonType	Each wagon model belongs to one wagon type.
WagonModel – Gauge	A wagon model can support one or more gauges, each gauge can apply to multiple wagon models.
Locomotive – LocomotiveModel	Each locomotive has one model type.
Train – Line (PlannedRoute)	A train can be planned to run on multiple lines, each line can be included in the planned routes of multiple trains.
Train – Line (ActualRoute)	A train can actually run on multiple lines during execution, each line can be used by multiple trains in their actual routes.
Train – Locomotive	A train uses one or more locomotives and a locomotive can serve different trains over time.
Train – Freight	A train can carry many freight records and each freight is assigned to one train
Freight – Wagon	Freight is loaded into wagons and a wagon can carry multiple freight items.
Wagon_Freight – FreightWagonUnload	Records when a specific wagon–freight assignment is unloaded.
LocomotiveModel – Diesel_locomotive / Electric_locomotive	A locomotive model can be either diesel or electric, not both.
LocomotiveModel – Gauge	A locomotive model can support one or more gauges, each gauge can apply to multiple locomotive models.
Operator – Locomotive / Wagon / Train	An operator manages one or more locomotives, wagons and trains.

Acronyms:

Acronym	Meaning
PK	Primary Key
FK	Foreign Key
ID	Identifier
VAT	Value Added Tax (used in VAT Number for business identification)
DBMS	Database Management System
Y/N	Yes or No (used for attributes like Electrified)

Summary of Basic Oracle Rules

1. Table Creation (DDL Rules)

We created all tables using Oracle's Data Definition Language. Each table was defined with clearly specified data types such as NUMBER for numeric values, VARCHAR2 for text fields and DATE for time-related attributes. We also assigned a primary key to every table to uniquely identify each record. Tables that depended on others included foreign keys referencing the primary key of another table, making sure that the relationships represented in the ER diagram were correctly applied in the database.

Many-to-many relationships were implemented using separate bridge tables containing two foreign keys, both of which together formed the primary key. This ensured proper relational structure and prevented duplicate combinations.

2. Data Insertion (DML Rules)

We inserted data using Oracle's Data Manipulation Language in a structured and consistent manner. Each INSERT operation followed the order of columns defined in the table. Text values were written using single quotes, while numeric values were inserted directly. During insertion, we always made sure that foreign key values already existed in the referenced table to avoid constraint violations. Insertions were performed only after the whole database structure was successfully created.

3. Updating and Deleting Data

For any adjustments, we used simple update and delete operations. Updates were applied only to specific rows by specifying appropriate conditions to avoid modifying unintended records. Deletions were also condition-based and carried out only when necessary.

4. Relationship Management

To maintain consistency with the ER diagram, all relationships were strictly implemented in Oracle using foreign keys. One-to-many relationships used a single foreign key in the child table, while many-to-many relationships were represented through dedicated bridge tables. These constraints prevented invalid data entry and ensured that all records respected the logical model of the system.

5. Naming Conventions and Structure

We followed a consistent naming style throughout the database. Primary keys used simple conventions such as "ID", "Number" or similar structured names. Foreign keys carried the same name as the referenced attribute, which made relationships easy to understand and trace. Bridge tables were named using a combination of the two linked tables, keeping the overall schema organized and intuitive.

6. Executing SQL in Oracle

All SQL commands were executed through Oracle SQL Developer. Tables were created in a specific sequence: first independent base tables, followed by related tables and finally bridge tables. After the structure was ready and all data was inserted, transactions were confirmed using Oracle's commit system to save the changes permanently.

7. Constraints Used

We applied standard Oracle constraints to control data quality:

- Primary keys ensured that each row was unique.
- Foreign keys enforced valid relationships.
- UNIQUE constraints prevented duplicated values in certain fields.
- NOT NULL constraints required essential fields to always contain data.

These constraints collectively ensured the reliability and integrity of the entire database.

Overall, we used Oracle in a structured and organized way by following standard SQL rules, applying consistent naming and relational integrity constraints, inserting data carefully and maintaining a clean execution order. This approach ensured that the implemented database matched the ER diagram and behaved as intended.